

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 3 – Vulnerability Detection**

Course Code:	CSA5803	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO3 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3
Weightage:	40% of CIA	Total Marks:	100
CO Statement:	3 Vulnerability Detection, Severity Analysis, Security Verification Workflow Simulation and Performance Evaluation		
Student Name:	Syed Saniya Iqbal	Reg. No.:	192465043
Section / Batch:	CSE-Cybersecurity	Date:	14/8/26

Instructions to Students

- Perform vulnerability detection and classify the identified vulnerabilities based on their severity levels.
- Conduct severity analysis using appropriate metrics such as CVSS and document the impact of each vulnerability.
- Design and simulate a Security Verification Workflow covering detection, remediation, re-testing, and verification.
- Evaluate security and system performance before and after mitigation, and include relevant results, screenshots, and observations.

Question Type	Focus Area	Questions	Marks
Vulnerability Detection	Conceptual Understanding	Q1	Q1 –100
GRAND TOTAL:		100 Marks	

Code of Conduct

I **Syed Saniya Iqbal(192465043)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **Syed Saniya Iqbal**

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

CLOUD-BASED SECURE DOCUMENT MANAGEMENT SYSTEM WITH CONTINUOUS SECURITY MONITORING

Five Core Security Modules

This document defines the five core modules of the proposed Cloud-Based Secure Document Management System with Continuous Security Monitoring. The system combines secure document management with continuous security assessment, detection, analysis, verification, and performance evaluation.

Module 1: Vulnerability Assessment

The Vulnerability Assessment module identifies security weaknesses that could affect the confidentiality, integrity, and availability of documents and the application.

Objectives

- Identify vulnerabilities in the document management application.
- Assess authentication and access-control weaknesses.
- Check insecure file-upload and file-access conditions.
- Identify weak security configurations and exposed services.
- Assign severity levels such as Critical, High, Medium, and Low.
- Generate vulnerability findings and remediation recommendations.

Module Workflow

Application and configuration data → Vulnerability Scanner → Vulnerability Findings → Severity Classification → Risk Information → Security Report

Module 2: Detection

The Detection module continuously monitors application and user activities to identify suspicious, abnormal, or potentially malicious behavior.

Objectives

- Monitor login and authentication activities.
- Monitor document upload, download, modification, and deletion events.
- Detect repeated failed logins and unusual access patterns.
- Detect suspicious file activities and unauthorized access attempts.
- Generate security alerts when predefined detection conditions are met.
- Record detected events for further analysis.

Module Workflow

Application/User Events → Monitoring Engine → Detection Rules → Event Correlation → Alert Generation → Security Event Log

Module 3: Security Analysis

The Security Analysis module examines detected events and related evidence to determine the severity, impact, and risk associated with a security event.

Objectives

- Collect logs and security events from the application.
- Correlate related events to identify attack patterns.
- Analyze user, document, authentication, and system activity.
- Classify events according to security severity.
- Calculate or assign a risk level.
- Provide analysis results to the security verification workflow.

Module Workflow

Security Alerts → Log Collection → Event Correlation → Security Analysis → Severity/Risk Assessment → Analysis Report

Module 4: Security Verification Workflow Simulation

The Security Verification Workflow Simulation demonstrates the complete process of verifying whether a detected security alert represents a true security incident or a false positive.

Objectives

- Receive a security alert from the detection module.
- Collect supporting logs, authentication records, and related events.
- Verify whether the activity is legitimate or suspicious.
- Classify the alert as True Positive or False Positive.
- For a true threat, initiate a simulated security response.
- Apply simulated remediation or mitigation.
- Re-scan or re-check the affected component to verify that the issue has been resolved.
- Record the complete verification workflow for reporting.

Module Workflow

Detection Alert → Evidence Collection → Security Verification → True Positive / False Positive → Response → Remediation → Re-Verification → Final Status

Module 5: Performance Evaluation

The Performance Evaluation module measures how effectively the security monitoring and verification system detects and handles security events.

Objectives

- Measure detection accuracy.
- Calculate detection rate.
- Measure response and verification time.
- Measure false-positive and false-negative rates where applicable.
- Evaluate vulnerability detection performance.
- Compare security performance before and after remediation.
- Generate performance statistics, charts, and an overall evaluation.

Module Workflow

Security Test Data → Performance Metrics → Calculation → Comparison → Charts/Statistics → Performance Evaluation Report

Combined Security Workflow

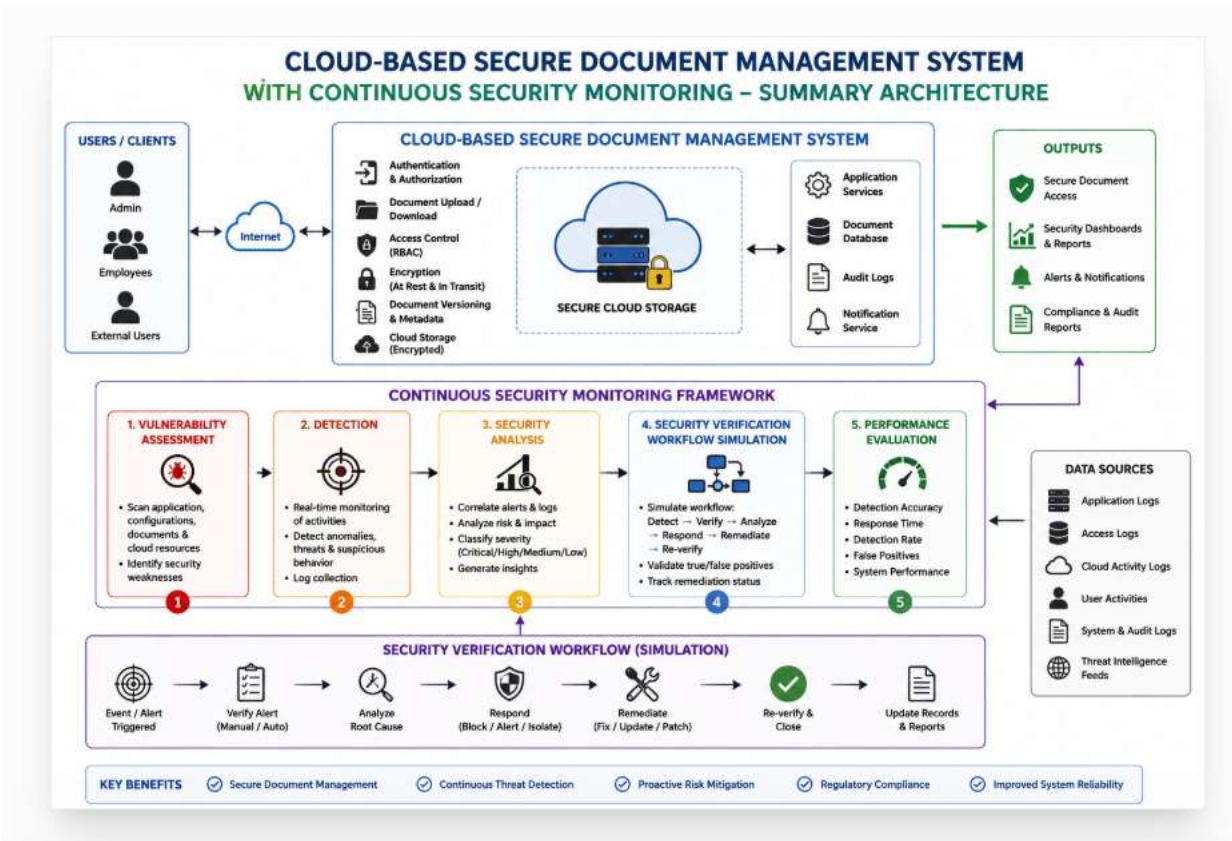
The five modules operate as a continuous security lifecycle. Vulnerabilities are assessed, security events are detected, alerts are analyzed, suspicious events are verified, and the overall effectiveness of the system is evaluated.

1. User / Application Activity
2. Secure Document Management
3. Vulnerability Assessment
4. Continuous Security Detection
5. Security Analysis
6. Security Verification Workflow
7. True Positive → Simulated Response → Remediation → Re-Verification
8. False Positive → Close Alert

- 9. Performance Evaluation
- 10. Final Security Report

Module Summary

Module	Main Purpose	Main Output
Vulnerability Assessment	Identify and classify security weaknesses	Vulnerability report
Detection	Continuously identify suspicious activities	Security alerts
Security Analysis	Analyze events and determine risk	Risk/analysis report
Security Verification Workflow	Verify alerts and simulate response/remediation	Verified incident status
Performance Evaluation	Measure effectiveness of the security system	Performance report



Conclusion

Together, these five modules transform the project from a basic cloud document storage application into a security-focused document management and continuous monitoring system. The workflow supports vulnerability identification, continuous detection, security analysis, verification of alerts, simulated remediation, and quantitative performance evaluation.